

Document checks covered text extraction, embedded citation links and page margins throughout the PDF. Visual inspection sampled seven pages, including the opening page, tables, proof sketch, Streams analysis and source record.

Source reading map

The following reading order groups the papers by the decision they inform. Links near the analysis support the corresponding claims; these publication details provide provenance.

- **Scheduling:** Matthew C. Loring, Mark Marron and Daan Leijen, *Semantics of Asynchronous JavaScript*, Microsoft Research technical report, 26 July 2017 / DLS 2017. Use the parameterized scheduler and stated abstraction limits.
- **Equivalence:** Li-yao Xia and coauthors, *Interaction Trees*, POPL 2020; Nicolas Chappe, Paul He, Ludovic Henrio, Yannick Zakowski and Steve Zdancewic, *Choice Trees*, arXiv version 1, 2022; Irene Yoon, Yannick Zakowski and Steve Zdancewic, *Formal Reasoning about Layered Monadic Interpreters*, ICFP 2022. Read weak equivalence, internal branching and layer-specific laws together.
- **Errors:** Gordon Plotkin and Matija Pretnar, *Handling Algebraic Effects*, LMCS 2013. Pair it with Web IDL's distinction between exception objects, throwing and promise rejection.
- **Rows:** Wouter Swierstra, *Data Types a la Carte*, JFP 2008; Oleg Kiselyov and Hiromi Ishii, *Freer Monads, More Extensible Effects*, Haskell 2015; Daan Leijen, *Koka: Programming with Row-Polymorphic Effect Types*, 2014; Ningning Xie and Daan Leijen, *Generalized Evidence Passing for Effect Handlers*, ICFP 2021. Separate inclusion, removal, duplicate labels and runtime evidence.
- **Scoped syntax:** Nicolas Wu, Tom Schrijvers and Ralf Hinze, *Effect Handlers in Scope*, Haskell 2014; Maciej Pirog, Tom Schrijvers, Nicolas Wu and Mauro Jaskelioff, *Syntax and Semantics for Operations with Scopes*, LICS 2018; Roger Bosman, Birthe van den Berg, Wenhao Tang and Tom Schrijvers, *A Calculus for Scoped Effects and Handlers*, LMCS 2024.
- **Deferred and modular elaboration:** Birthe van den Berg, Tom Schrijvers, Casper Bach-Poulsen and Nicolas Wu, *Latent Effects for Reusable Language Components*, APLAS 2021; Casper Bach Poulsen and Cas van der Rest, *Hefty Algebras*, POPL 2023; Birthe van den Berg and Tom Schrijvers, *A Framework for Higher-Order Effects and Handlers*, 2023 preprint.
- **Machines and compilation:** Olivier Danvy and Lasse R. Nielsen, *Defunctionalization at Work*, BRICS 2001; Daan Leijen, *Type Directed Compilation of Row-Typed Algebraic Effects*, POPL 2017; Daniel Hillerstrom and Sam Lindley, *Liberating Effects with Rows and Handlers*, TyDe 2016; KC Sivaramakrishnan and coauthors, *Retrofitting Effect Handlers onto OCaml*, PLDI 2021; Ohad Kammar, Sam Lindley and Nicolas Oury, *Handlers in Action*, ICFP 2013, as a broader operational introduction.
- **Composition:** Zhixuan Yang and Nicolas Wu, *Reasoning about Effect Interaction by Fusion*, ICFP 2021. Apply its sufficient conditions to chosen handler clauses rather than assuming a general stream fusion law.

The remaining uncertainty is concentrated in formalization and admission: exact M2 events, cause projection, instance identity, callback semantics, fairness premises and graph reducibility. The research supplies evidence and falsifiers for deciding those points; it does not turn the proposed decisions into established theorems.